Modelling a 3D printer

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October 17, 2026

0 Introduction

3D Printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Different than Cartesian designs, where one motor drives one axis, here each motor simultaneously changes the x and y position of the tool head (see Figure 1). This results in a more complex kinematic model, but allows a more sturdy casing and faster printing speeds. Our task is to model and simulate the system.

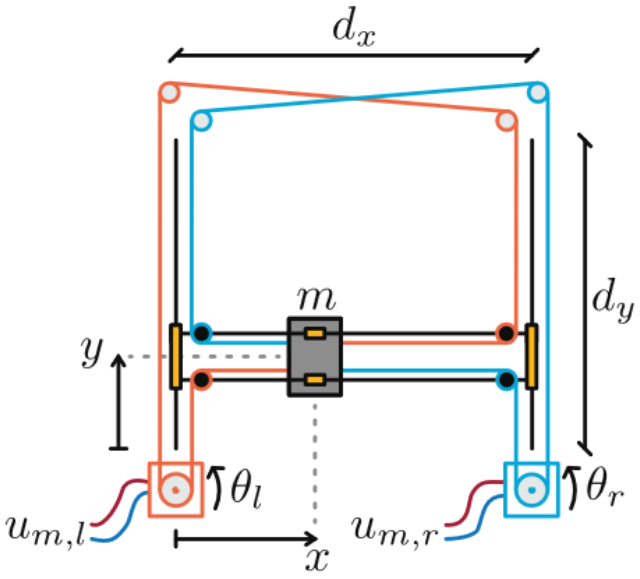


Figure 1: Schematic of the Core XY mechanism.

0.1 Symbol table

Table 1: Symbols and parameter values.

Description	Symbol	Value
Motor angle left	θ_l	–
Motor voltage left	$u_{m,l}$	–
Motor current left	$i_{m,l}$	–
Motor radius	r_m	$11\,\mathrm{mm}$
Motor induction	L_m	$10\,\mathrm{mH}$
Motor resistor	R_m	$90\,\Omega$
Motor inertia	J_m	$8 \times 10^{-6}\,\mathrm{kg\,m^2}$
Motor constant	K_m	$0.07\,\mathrm{N\,m/A}$
Tool head position	x, y	–
Tool head mass	m	$60\,\mathrm{g}$
Nozzle voltage	u_n	–
Nozzle current	i_n	–
Nozzle capacitor	C_n	–
Nozzle thermal capacity	$C_{t,n}$	–
Nozzle thermal resistance	$R_{t,n}$	$9\,\mathrm{K/W}$
Nozzle resistor	R_n	$8\,\Omega$
Nozzle temperature	T_n	$240\,^\circ\mathrm{C}$
Environmental temperature	T_e	$300\,\mathrm{K}$
Bearing friction rolling	μ_r	$0.003\,\mathrm{N\,s/m}$
Bearing friction linear x-direction	$\mu_{l,x}$	$0.002\,\mathrm{N\,s/m}$
Bearing friction linear y-direction	$\mu_{l,y}$	$0.001\,\mathrm{N\,s/m}$
Bearing radius	r_b	$3\,\mathrm{mm}$
Print bed temperature	T_b	$105\,^\circ\mathrm{C}$
Print area dimension x	d_x	$29\,\mathrm{cm}$
Print area dimension y	d_y	$29\,\mathrm{cm}$

1 Physical Modelling

1.1 Kinematics

First task is to derive the direct kinematics of the Core XY mechanism. One fact that slightly complcates this is the fact that rotating one motor will move the tool head in both x and y directions. By removing the right motor, we can see that by moving the tool head in the positive x direction **or** in the positive y direction, the left motor will rotate in the positive direction. We can draw similar conclusions for the right motor, for which we get the following equatons:

$$\begin{aligned} \Delta A &= \Delta x + \Delta y \\ \Delta B &= \Delta x - \Delta y \\ \Delta x &= \frac{\Delta A + \Delta B}{2} \\ \Delta y &= \frac{\Delta A - \Delta B}{2} \end{aligned} \tag{1}$$

With ΔA and ΔB being the length of belt moved by the motors, and Δx and Δy being the change in position of the tool head.

$$\begin{aligned} \Delta A &= r_m \, \theta_L \\ \Delta B &= r_m \, \theta_R \end{aligned} \tag{2}$$

If we assume that in the beginning the motors are at $\theta_L = \theta_R = 0$ and the cart is at $(0, 0)$, we get the following equations:

$$\begin{aligned} x(\theta_L, \theta_R) &= \frac{r_m \theta_L + r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L + \theta_R\right) \\ y(\theta_L, \theta_R) &= \frac{r_m \theta_L - r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L - \theta_R\right) \end{aligned} \tag{3}$$

$$\begin{aligned} \dot{x}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L + \dot{\theta}_R\right) \\ \dot{y}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L - \dot{\theta}_R\right) \end{aligned} \tag{1}$$

1.2 Dynamics

In next part we model the dynamics of the 3D printer system by applying the Euler-Lagrange equation. Additionally, we will split the problem to a mechanical and an electrical one, with the angular velocity of the motor shafts connecting them.

Because of that, we choose generalized coordinates:

$$q = \begin{bmatrix} x \\ y \\ q_A \\ q_B \end{bmatrix} \qquad \dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ i_A \\ i_B \end{bmatrix}$$

We can also then split them among our two problems:

$$\dot{q}_m = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \text{ (mechanical part)}, \qquad \dot{q}_e = \begin{bmatrix} i_A \\ i_B \end{bmatrix} \text{ (electrical part)}$$

1.2.1 Mechanical part

Kinetic energy

Fistly, we derive the kinetic energy terms. In the *mechanical* part, the only mass is the tool head. With that we have:

$$K_m = \tfrac{1}{2} m(\dot{x}^2 + \dot{y}^2) = \tfrac{1}{2}\, \dot{q}_m^\top M\, \dot{q}_m \quad with \quad M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}$$

As the whole system is planar, we can see that there is no potential energy terms as it will be constant.

Dissipative

Secondly, we find the dissipative energy terms. Each motor has 4 rotational bearings, and 4 linear bearings for the whole system. We define r_b as the bearing radius, later it will be kept the same as the motor radius for simplcity. From (eq 4), we can get transformations (7) with trivial time derivatives.

$$\theta_L = \frac{x+y}{r_b}, \qquad \theta_R = \frac{x-y}{r_b}$$

Then, for each side we get rotational dissipative energy:

$$D_L = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_L^2\, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2\right]$$
$$D_R = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_R^2\, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 - 2\dot{x}\dot{y} + \dot{y}^2\right]$$

Then simply combining them:

$$\boxed{D_{\text{rot}} = 4\, \frac{\mu_r}{r_b^2} \left[\dot{x}^2 + \dot{y}^2\right]}$$

Similarly, we get the linear dissipative energy:

$$D_{\text{lin}} = \tfrac{1}{2}\mu_{L,x}\, \dot{x}^2 + \tfrac{1}{2}\mu_{L,y}\, \dot{y}^2$$

Then, combining the terms we get:

$$D_m = \left[4\, \frac{\mu_r}{r_b^2} + \tfrac{1}{2}\mu_{L,x}\right] \dot{x}^2 + \left[4\, \frac{\mu_r}{r_b^2} + \tfrac{1}{2}\mu_{L,y}\right] \dot{y}^2$$

1.2.2 Electrical part

2 Finite State Machine

3 Hybrid System

References